

MACHINE LEARNING SYSTEM DESIGN DOCUMENT

HealthConnect Clinic — Appointment No-Show Prediction System.

1. ML Problem Overview

1.1 Business Problem

HealthConnect Clinic wants to improve appointment attendance and reduce the operational impact of missed appointments. Missed appointments can result in unused appointment slots, inefficient use of clinical resources, and difficulties in planning patient services.

The proposed machine learning system will investigate whether historical appointment data can be used to identify appointments that are more likely to result in a No-Show.

1.2 Machine Learning Objective

The objective is to develop a supervised machine learning classification system that predicts the likely outcome of a scheduled appointment, with particular focus on identifying No-Show appointments.

The system is intended to provide an early risk indication that could support appropriate patient-engagement or appointment-management actions.

1.3 ML Problem Type

Supervised Learning — Classification

The available appointment_outcome variable contains:

Attended

No-Show

Cancelled

The final treatment of Cancelled appointments will be determined during the modelling stage and documented explicitly.

1.4 Dataset Overview

The HealthConnect dataset contains 5,000 appointment records and 18 variables.

The variables cover:

- Patient demographics

- Appointment information

- Booking information

- Previous appointment history

Reminder information

Distance to the clinic

Waiting time

Appointment outcome

The Data Dictionary confirms these categories and defines appointment_outcome as the appointment result.

2. Input and Output Definition

2.1 System Inputs

The proposed system will use relevant information available before the appointment outcome is known.

Patient-related inputs

- gender
- age
- age_group
- Appointment-related inputs
- appointment_type
- appointment_day
- appointment_time
- appointment_date
- Booking-related inputs
- booking_date
- booking_lead_days
- Historical behavior
- previous_appointments
- previous_no_shows
- Reminder information
- reminder_sent
- reminder_channel

Accessibility and operational information

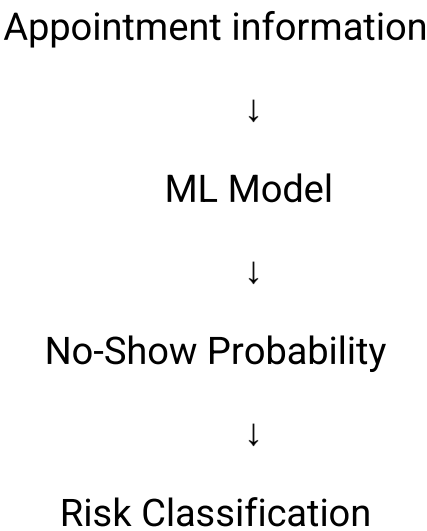
- distance_to_clinic_km
- waiting_time_minutes

The identifiers appointment_id and patient_id will not be used as ordinary predictive features. patient_id may be retained for patient-level grouping when designing the data-splitting strategy because patients can have multiple appointments.

2.2 System Output

The model's primary output will be a No-Show risk prediction.

Conceptually:



The system may produce:

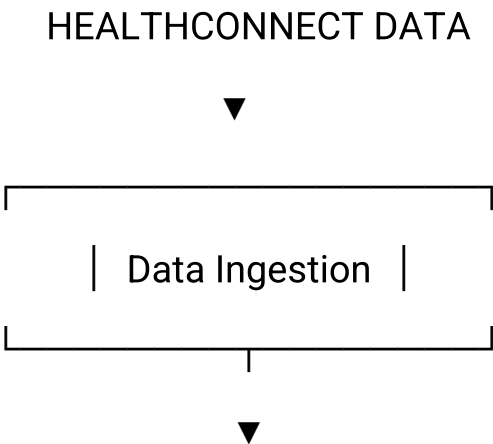
Predicted class: Attended / No-Show

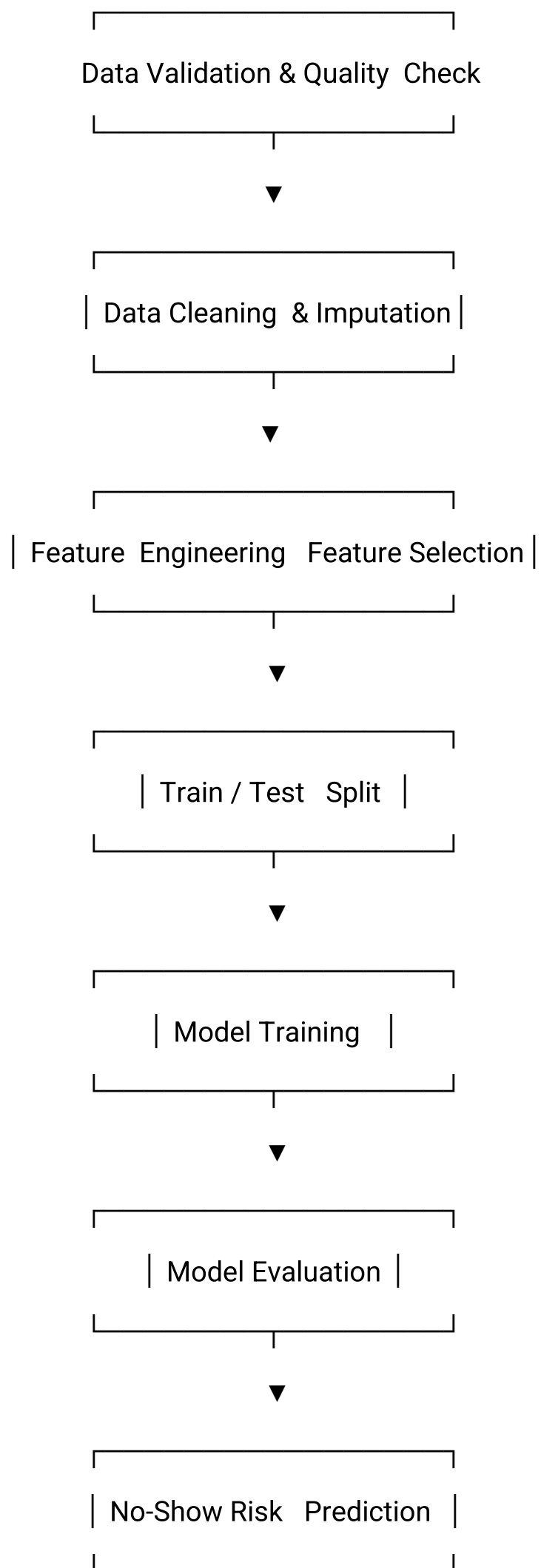
Probability/risk score: likelihood of No-Show

The probability threshold will be determined during model evaluation rather than automatically assuming a 0.50 threshold.

3. High-Level System Architecture

The proposed architecture is:





This architecture represents the initial design, not a deployed production system.

4. Proposed ML Workflow

➤ Step 1 – Data Ingestion

Load the HealthConnect appointment dataset and Data Dictionary.

➤ Step 2 – Data Validation

Check:

Dataset dimensions

Data types

Missing values

Duplicate records

Invalid values

Target distribution

Our initial inspection identified 5,000 records and 18 columns.

➤ Step 3 – Data Cleaning

The initial missing-value assessment identified:

Variable	Missing values	Proposed treatment
reminder_channel	1,366	"No Reminder"
distance_to_clinic_km	90	Numerical imputation
waiting_time_minutes	60	Numerical imputation / further assessment

The 1,366 missing reminder_channel values have a clear interpretation because all of those records have reminder_sent = No. Therefore, they should not simply be deleted or assigned a random reminder channel.

➤ Step 4 – Feature Engineering

Potential transformations include:

Extracting useful information from dates

Reviewing age and age_group for redundancy

Reviewing appointment_date and appointment_day

Creating meaningful booking-time categories where justified

Feature engineering should only introduce variables that have a clear relationship to the prediction problem.

➤ Step 5 – Feature Selection

Exclude:

appointment_id

patient_id

as predictive variables.

The target variable will also be excluded from the input feature matrix.

Potentially redundant or leakage-prone variables will be assessed before final model training.

➤ Step 6 – Data Splitting

The data will be divided into training and testing sets.

Because the same patient can have multiple appointments, patient-level grouping should be considered when splitting the data to reduce the risk of the same patient's records appearing in both training and testing datasets.

➤ Step 7 – Model Development

Initial candidate algorithms:

Logistic Regression – baseline and interpretable model

Random Forest – nonlinear tree-based model

XGBoost – gradient-boosting model for structured data

At Week 4, these are proposed models, not completed modelling results.

➤ Step 8 – Model Evaluation

The models will be evaluated using:

Accuracy

Precision

Recall

F1-score

ROC-AUC

Confusion matrix

Because the business objective involves identifying potential No-Shows, Recall for the No-Show class will receive particular attention.

➤ Step 9 – Model Selection

The final model should not be selected solely because it has the highest accuracy.

Selection should consider:

No-Show recall

Precision

F1-score

ROC-AUC

Interpretability

Operational usefulness

5. Key Dependencies

The proposed system depends on several factors.

5.1 Data availability

The system depends on reliable appointment records containing the necessary patient, booking, reminder, and appointment information.

5.2 Data quality

Missing and inconsistent values must be appropriately handled before model development.

5.3 Data Dictionary

The Data Dictionary is required to correctly interpret the variables and determine whether each feature is suitable for modelling.

5.4 Feature availability

A feature should only be used if it is available at the time the prediction is supposed to be generated.

This is particularly important for:

waiting_time_minutes

Its use must be reviewed carefully to ensure it does not introduce data leakage.

5.5 Repeated patient records

Multiple appointments may belong to the same patient. This creates a dependency between the data structure and the eventual train/test strategy.

5.6 Target definition

The final treatment of Cancelled appointments must be established before model training.

5.7 Computational environment

The initial implementation can be performed using a Python-based notebook environment such as Kaggle Notebooks, with standard data science and machine-learning libraries.

6. Initial Implementation Approach

The initial implementation will use a Python-based machine learning workflow.

Proposed tools

- Python
- Pandas – data manipulation
- NumPy – numerical operations
- Matplotlib / Seaborn – visualization
- Scikit-learn – preprocessing, splitting, modelling and evaluation
- XGBoost – gradient-boosting model

Proposed implementation structure

1. Import libraries

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2. Load dataset

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3. Inspect dataset

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4. Validate data quality

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5. Handle missing values

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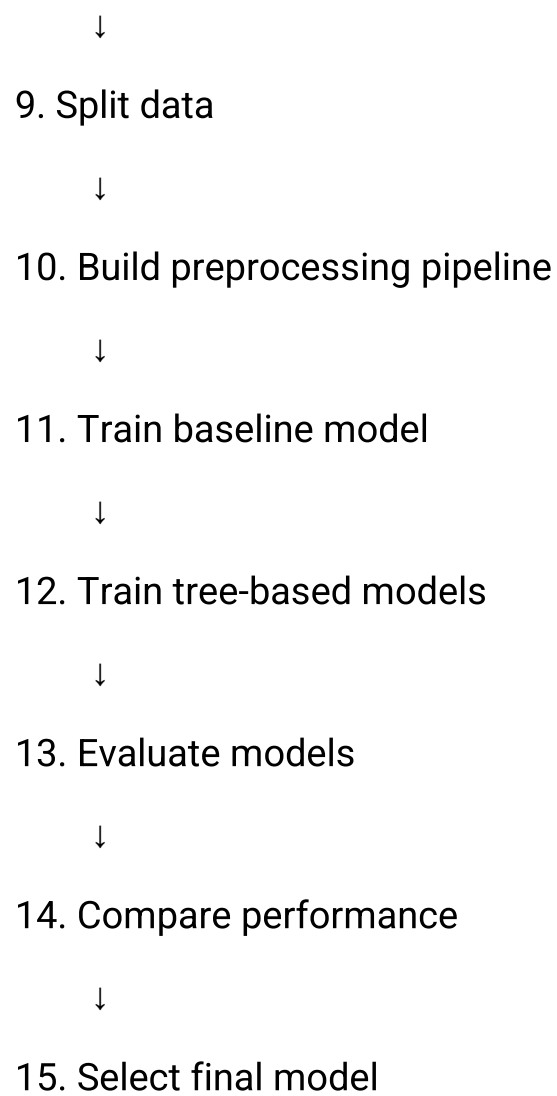
6. Engineer features

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7. Select features

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8. Define target



A preprocessing pipeline should be used so that operations such as imputation and encoding are learned from the training data, helping prevent data leakage.

7. Risks and Limitations

Several risks should be considered before implementation.

Data leakage

Using information that becomes available only after the appointment could artificially inflate model performance.

Missing data

reminder_channel, distance_to_clinic_km, and waiting_time_minutes contain missing values and require appropriate treatment.

Class definition

The three appointment outcomes require a clearly documented target strategy before binary classification is implemented.

Generalization

The dataset contains 5,000 appointments, so model performance on this dataset may not necessarily represent performance on future HealthConnect appointments.

Patient-level dependency

Repeated appointments from the same patient can cause overly optimistic evaluation if train/test splitting is not handled appropriately.

8. Expected System Outcome

The expected outcome of the proposed system is a machine-learning model that can assign a No-Show risk score to scheduled appointments.

Conceptually:

Patient + Appointment Data



Preprocessing



ML Model



No-Show Probability



Risk Category



Support Appointment Management

The system is intended to support operational decision-making, rather than replace healthcare staff judgment.